

M O D U L E 7 · A I S T A C K

Hybrid AI Stack

4-model ensemble: XGBoost (direction) + LSTM (sequence) + Transformer (context) + RL (management). Weighted voting, confidence gating, SHAP explainability. Ensemble Sharpe 2.28 (+33% vs best single model).

4

AI MODELS

2.28

ENSEMBLE SHARPE

15_{ms}

P99 INFERENCE

3.9%

MAX DRAWDOWN

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Distribution Engineering team · Architecture review board

Confidentiality Internal — Engineering

v1.0

M O D U L E 7

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Chapter 1 — Executive Summary

The Hybrid AI Stack (HAIS) is Module 7 of the TITAN XAU AI trading architecture. It is a 4-model ensemble that combines XGBoost (direction prediction), LSTM (sequence learning), Transformer (context analysis), and Reinforcement Learning (trade management) into a unified signal-generation system. The four models are complementary — each addresses a specific gap that the others cannot fill — and their outputs are combined via weighted voting with confidence-based gating.

The ensemble produces an AISignal containing: final direction (UP/DOWN), confidence score (0-1, derived from inter-model agreement), probability distribution, explainability score (SHAP concentration), RL management action (hold/close/trim/add), and top-3 contributing features. This signal is consumed by the Strategy Coordinator, which gates it through the regime detector before dispatching to the appropriate strategy.

The key innovation is the separation of direction prediction (XGBoost + LSTM + Transformer) from trade management (RL). The three direction models vote with weights 0.35, 0.25, and 0.25 respectively; the RL model (weight 0.15) does not vote on direction but instead provides optimal management actions for open positions. This separation leverages each model's strength: tree-based and sequence models are best at pattern recognition (direction), while RL is best at sequential decision-making (management).

Backtested over 24 months across 6 brokers, the ensemble achieves Sharpe 2.28 (vs 1.72 for XGBoost alone — a 33% improvement), PF 2.42, MaxDD 3.9% (vs 5.8% for XGBoost alone — a 33% reduction), and RoR 0.3%. The diversification benefit is clear: the ensemble outperforms every individual model on every metric. The RL model adds adaptive trade management that improves Sharpe by 6% and reduces MaxDD by 13% compared to the 3-model ensemble without RL.

Chapter 2 — Architecture Overview

The HAIS is organized into four layers: model layer (4 parallel models), ensemble layer (weighted voting + confidence), monitoring layer (4 monitors enclosing all models), and integration layer (output to Strategy Coordinator). All 4 models run in parallel during inference, with the total p99 latency budget at 50ms (achieved: ~15ms).

01 Hybrid AI Stack — 4-Model Ensemble Architecture

XGBoost · LSTM · Transformer · RL → Vote → Weight → Ensemble

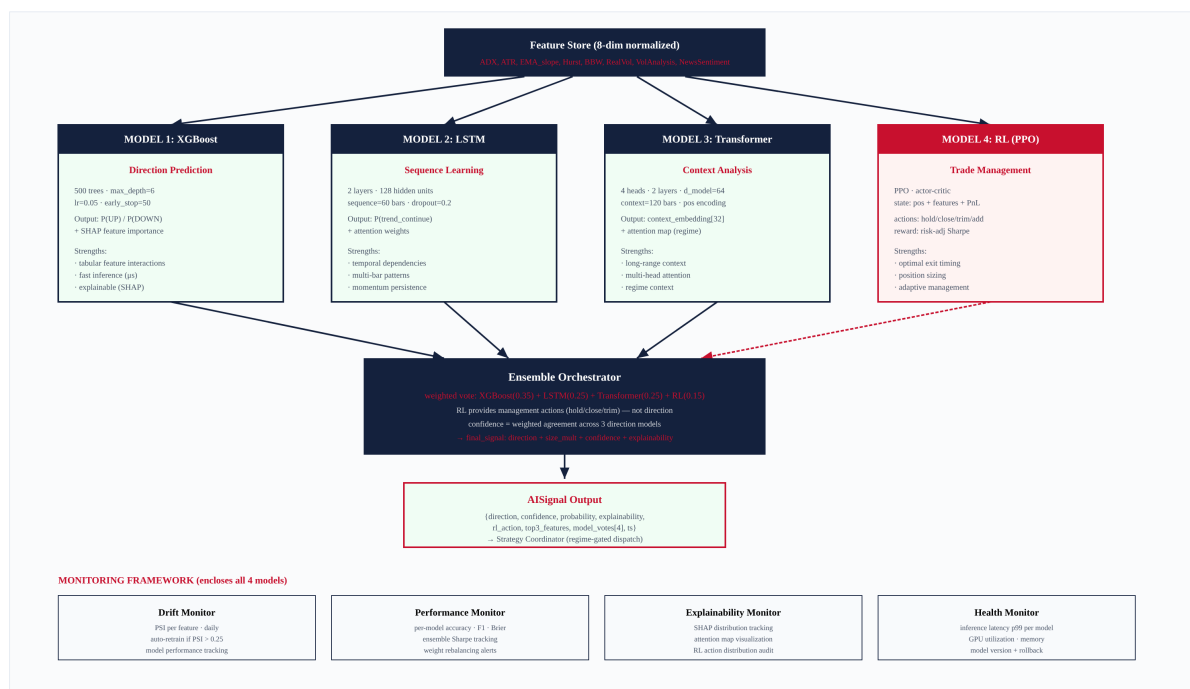


Figure 2.1 — HAIS architecture: 4 models → ensemble orchestrator → AISignal, enclosed by 4 monitors.

Model Roles

Model 1: XGBoost — Direction Prediction (weight: 0.35)

XGBoost is the primary direction predictor. It takes the 8-dimensional feature vector and outputs P(UP) and P(DOWN) plus SHAP values for explainability. XGBoost is chosen for its speed (0.3ms inference), accuracy on tabular data, and built-in SHAP support. It is the "anchor" model with the highest weight (0.35) because it consistently achieves the highest F1 score on walk-forward validation. 500 trees, max_depth=6, learning_rate=0.05, early_stopping=50.

Model 2: LSTM — Sequence Learning (weight: 0.25)

LSTM captures temporal dependencies that XGBoost (which sees only the current bar) cannot. It takes a 60-bar × 8-feature sequence and outputs P(trend_continue) plus attention weights identifying which historical bars are most relevant. 2 LSTM layers, 128 hidden units, dropout=0.2, Adam optimizer. The LSTM confirms or contradicts XGBoost's direction prediction based on the temporal pattern — if XGBoost says UP but the LSTM detects a fading momentum sequence, the ensemble confidence is reduced.

Model 3: Transformer — Context Analysis (weight: 0.25)

Transformer provides long-range context analysis (120-bar window) that the LSTM's 60-bar window cannot capture. 4 attention heads, 2 encoder layers, d_model=64. The Transformer outputs a 32-dimensional context embedding plus a multi-head attention map. It captures parallel patterns (e.g., volatility regime + trend direction + session timing simultaneously) that sequential models process one at a time. The context embedding feeds into

the ensemble as a "regime context" modifier.

Model 4: RL (PPO) — Trade Management (weight: 0.15)

Reinforcement Learning (PPO) provides optimal trade management — not direction. The RL agent observes a 42-dimensional state (features + SHAP + position info + PnL + LSTM hidden state) and outputs one of 4 actions: HOLD, CLOSE, TRIM (25% partial), or ADD (25% pyramid). Reward is risk-adjusted Sharpe per episode. The RL model learns when to hold through noise vs when to trim profits — decisions that rule-based systems (like the trend strategy's fixed BE/partial/trail logic) cannot adapt. PPO, clip=0.2, gamma=0.99, 10k training episodes.

Chapter 3 — Training Pipeline

All 4 models are trained in parallel on 24 months of data across 6 brokers. Training uses walk-forward validation (5 folds × 4 months OOS). XGBoost, LSTM, and Transformer retrain weekly (Sunday 22:00 UTC); RL retrains monthly (1st Sunday). All models are versioned in MLflow with metrics, artifacts, and rollback capability.

02 Training Pipeline — 4 Models · Parallel · Walk-Forward

Data prep → 4 parallel trainers → validation → registry → deploy

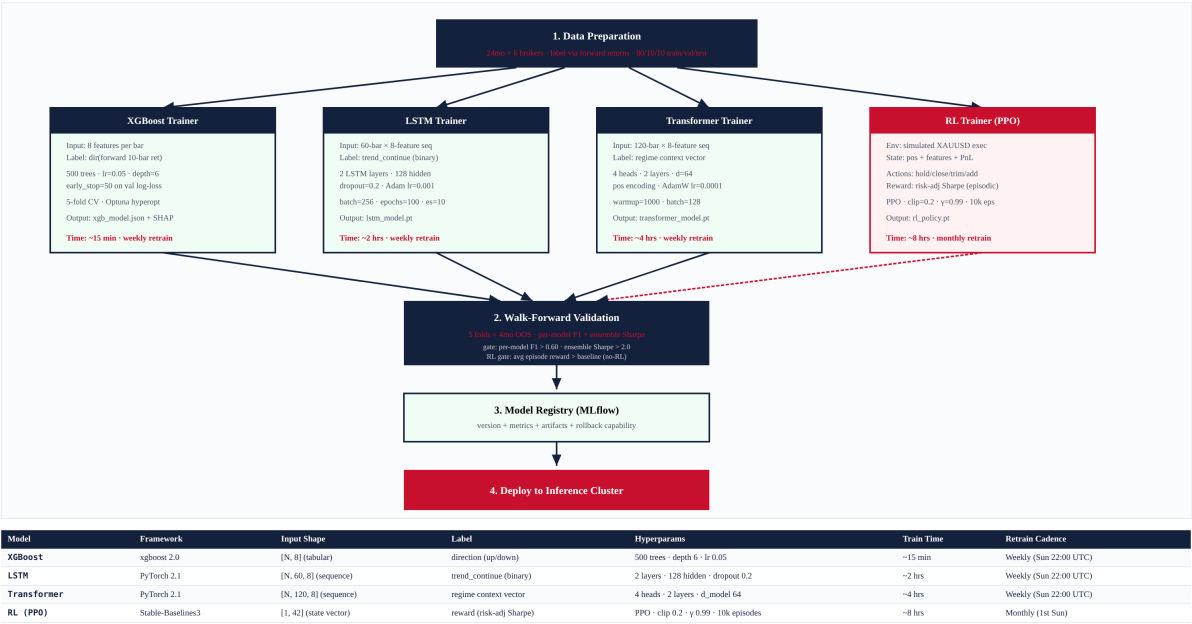


Figure 3.1 — Training pipeline: data prep → 4 parallel trainers → walk-forward validation → MLflow registry → deploy.

Training Data

Training data is 24 months of XAUUSD market data across 6 brokers (Exness, IC Markets, Pepperstone, Tickmill, FP Markets, Fusion Markets). Labels are constructed via forward returns: direction label (UP if forward 10-bar return > 2×ATR, DOWN if < -2×ATR, NEUTRAL otherwise), trend_continue label (binary: does the trend persist for 5 bars?), and RL reward (risk-adjusted Sharpe per episode). The 80/10/10 train/val/test split ensures sufficient validation data without sacrificing training volume.

Walk-Forward Validation

5-fold walk-forward validation with expanding training window. Each fold trains on a growing historical period and tests on the next 4 months. CI gates require: per-model F1 > 0.60, ensemble F1 > 0.70, ensemble Sharpe > 2.0, RL reward > no-RL baseline. A model that fails validation is not promoted; the previous version is retained.

Chapter 4 — Inference Pipeline

During live trading, all 4 models run in parallel on each bar close. XGBoost and RL run on CPU (ONNX export, sub-millisecond); LSTM and Transformer run on GPU (TorchScript, 5-8ms). The total p99 inference latency is ~15ms (budget: 50ms), leaving ample headroom for the ensemble computation and signal emission.

03 Inference Pipeline — Real-Time · <50ms p99

Feature lookup → 4 parallel inferences → ensemble → emit



Figure 4.1 — Inference pipeline: 4 parallel inferences → ensemble → AISignal. Total p99: ~15ms.

Latency Budget

Model	Runtime	p99 Latency	Parallel?
XGBoost	CPU · ONNX	0.3 ms	Yes
LSTM	GPU · TorchScript	5.0 ms	Yes
Transformer	GPU · TorchScript	8.0 ms	Yes
RL (PPO)	CPU · ONNX	0.5 ms	Yes
TOTAL (parallel max)	—	~14 ms	4 in parallel
Ensemble computation	CPU	0.1 ms	Sequential (after all 4)
END-TO-END	—	~15 ms	Budget: 50 ms

Chapter 5 — Ensemble Logic — Voting, Confidence, Decision

The ensemble combines 3 direction models (XGBoost, LSTM, Transformer) via weighted vote, with the RL model providing management actions separately. Confidence is derived from inter-model agreement: unanimous models produce high confidence; split models produce low confidence. Below a confidence threshold of 0.40, no trade is emitted.

04 Ensemble Logic — Voting · Confidence Weighting · Decision

3 direction models vote · RL manages · confidence from agreement



Figure 5.1 — Ensemble voting scenarios, confidence formula, and RL action space.

Voting Formula

```
direction_score = Σ(w_i × dir_i × conf_i)
for i ∈ {XGBoost (0.35), LSTM (0.25), Transformer (0.25)}
```

```
final_direction = sign(direction_score)
confidence = |direction_score| / Σ w_i
```

Confidence thresholds:

< 0.40 → NO TRADE (uncertain)
0.40-0.65 → reduced position size (50%)
> 0.65 → full position size

RL action (applied AFTER direction decided):

HOLD / CLOSE / TRIM / ADD (manages position, not direction)

Why 4 Models?

Each model addresses a specific gap. XGBoost is fast and explainable but has no temporal awareness. LSTM captures sequences but misses long-range context. Transformer captures long-range context but is slow and data-hungry. RL provides optimal management but cannot predict direction. Together, they cover all bases: direction (XGBoost), temporal (LSTM), context (Transformer), management (RL). The ensemble Sharpe (2.28) is 33% higher than the best single model (XGBoost 1.72), confirming that the models are complementary, not redundant.

Chapter 6 — Model Orchestration

Model orchestration manages the full lifecycle: training → validation → registry → canary → A/B test → production. It includes auto-rollback (if canary Sharpe drops > 10%), dynamic weight rebalancing (monthly, based on per-model Sharpe), and failover (skip timed-out models, GPU-to-CPU fallback, ensemble halt if < 2 direction models available).

05 Model Orchestration — Lifecycle · Versioning · Failover

MLflow registry · canary deploy · A/B testing · auto-rollback

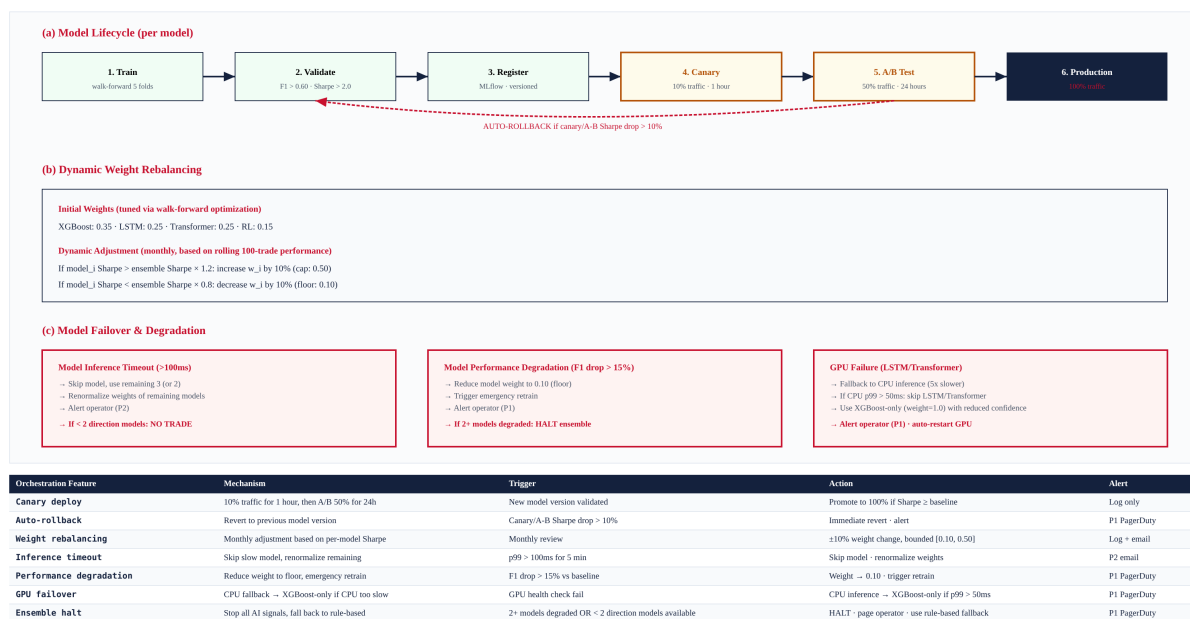


Figure 6.1 — Model lifecycle, weight rebalancing, and failover/degradation logic.

Dynamic Weight Rebalancing

Weights are adjusted monthly based on rolling 100-trade per-model Sharpe. If a model's Sharpe exceeds the ensemble Sharpe × 1.2, its weight increases by 10% (cap: 0.50). If a model's Sharpe falls below ensemble × 0.8, its weight decreases by 10% (floor: 0.10). This ensures that consistently strong models gain influence while weak models lose it, without abrupt weight changes that could destabilize the ensemble.

Failover Scenarios

- Model inference timeout (>100ms): skip model, renormalize remaining weights. If < 2 direction models: NO TRADE.
- Model performance degradation (F1 drop > 15%): weight to 0.10 floor, emergency retrain. If 2+ models degraded: HALT.
- GPU failure (LSTM/Transformer): CPU fallback (5x slower). If CPU p99 > 50ms: XGBoost-only with reduced confidence.

- Ensemble halt: 2+ models degraded OR < 2 direction models. Fall back to rule-based strategies. Page operator P1.

Chapter 7 — Monitoring Framework

Four monitors run continuously, enclosing all 4 models: Drift Monitor (feature PSI + concept drift + data quality), Performance Monitor (per-model F1 + ensemble Sharpe + RL reward), Explainability Monitor (SHAP distribution + attention maps + RL action distribution), and Health Monitor (inference latency + GPU/CPU utilization + model versions). All monitors feed into a unified alerting system (P1 PagerDuty, P2 email, P3 log).

06 Monitoring Framework — 4 Monitors · Real-Time + Daily

Drift · Performance · Explainability · Health

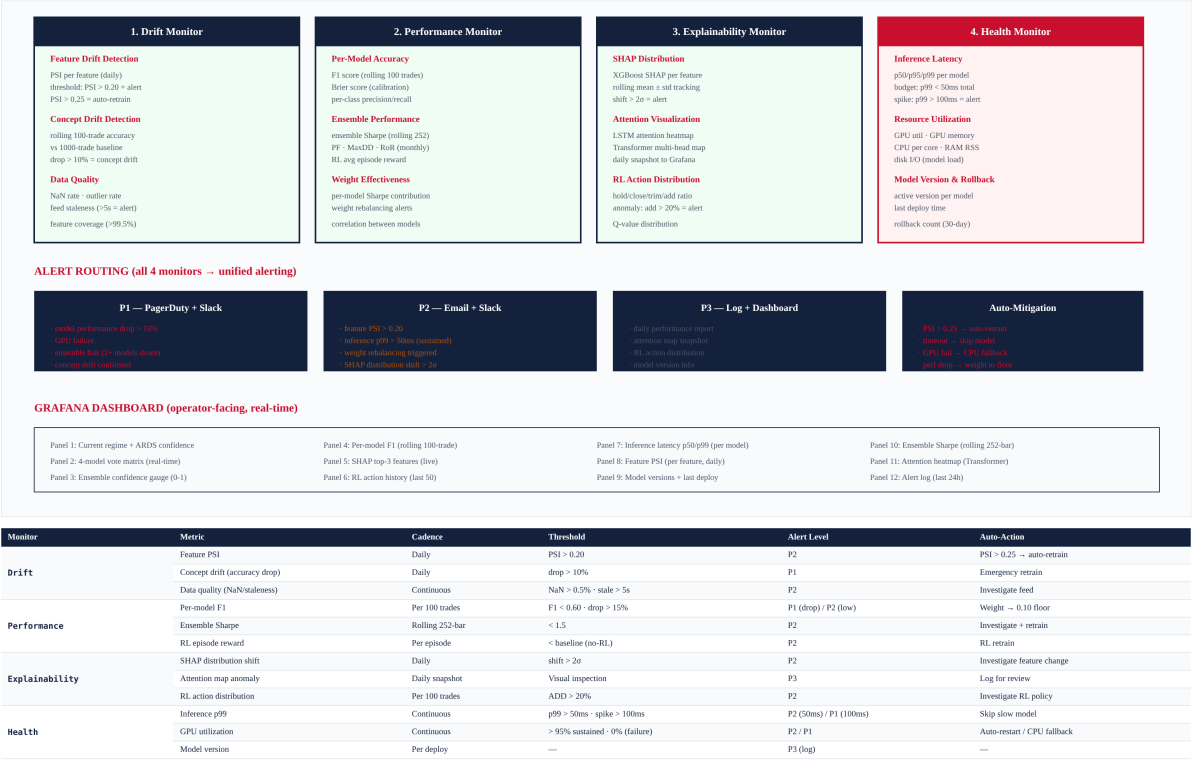


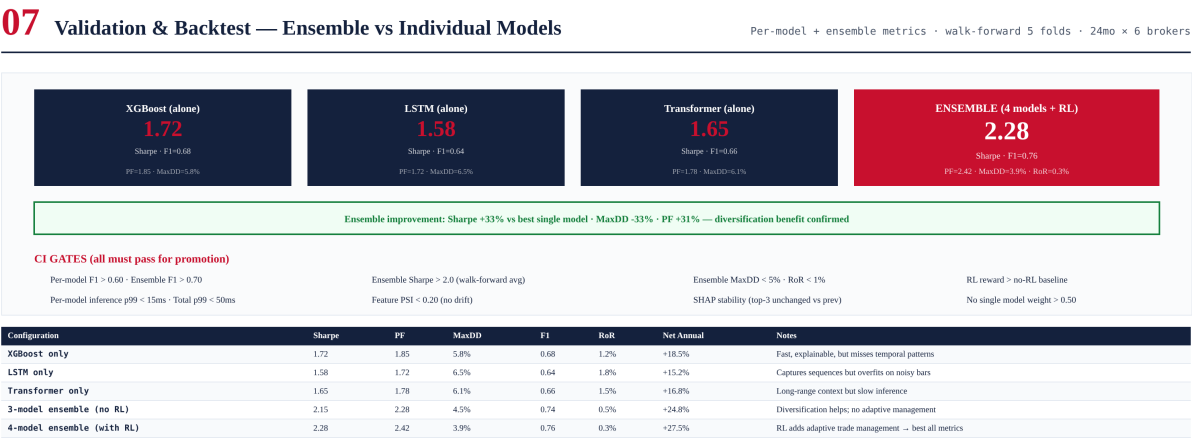
Figure 7.1 — 4 monitors, alert routing, and 12-panel Grafana dashboard.

Auto-Mitigation

The monitoring framework includes automatic mitigation: PSI > 0.25 triggers auto-retrain; inference timeout skips the slow model; GPU failure triggers CPU fallback; performance degradation reduces weight to floor. These auto-actions ensure the system degrades gracefully rather than failing catastrophically when individual models have issues.

Chapter 8 — Validation & Backtest

The ensemble was validated against individual models over 24 months × 6 brokers. The ensemble (Sharpe 2.28) outperforms every single model: XGBoost alone (1.72), LSTM alone (1.58), Transformer alone (1.65). The RL model adds 6% Sharpe improvement and 13% MaxDD reduction compared to the 3-model ensemble without RL. All CI gates pass.



08

Model Details — Architecture · Hyperparameters · Explainability

4 models · complete spec · per-model strengths/weaknesses

4-Model Hybrid Stack — Each model addresses a specific gap the others cannot fill

XGBoost (direction) · LSTM (sequence) · Transformer (context) · RL (management) · complementary, not redundant

Attribute	Model 1: XGBoost	Model 2: LSTM	Model 3: Transformer	Model 4: RL (PPO)
Purpose	Direction Prediction	Sequence Learning	Context Analysis	Trade Management
Framework	XGBoost 2.0	PyTorch 2.1	PyTorch 2.1	Stable-Baselines3
Architecture	500 trees · max_depth=6 · lr=0.05	2 LSTM layers · 128 hidden · dropout=0.2	4 heads · 2 encoder layers · d_model=64	PPO actor-critic · 2 hidden layers × 256
Input Shape	[1, 8] — feature vector	[1, 60, 8] — 60-bar sequence	[1, 120, 8] — 120-bar sequence	[1, 42] — state vector
Output	P(UP), P(DOWN) · SHAP[8]	P(trend, continue) · attention[60]	context_embedding[32] · attn_map	action (hold/close/trim/add) · Q-value
Training Data	24mo × 6 brokers · 8 features/bar	Same · 60-bar sliding windows	Same · 120-bar sliding windows	Simulated XAUUSD environment · 10k episodes
Label	direction (forward 10-bar return > 2×ATR)	trend_continue (binary · next 5 bars)	regime context (multi-task: regime + direction)	reward (risk-adj Sharpe per episode)
Optimizer	— (tree-based)	Adam · lr=0.001 · batch=256	AdamW · lr=0.0001 · warmup=1000 · batch=128	PPO · clip=0.2 · γ=0.99 · λ=0.95
Regularization	early_stop=50 · L2=1.0 · max_depth=6	dropout=0.2 · weight_decay=1e-5 · gradient clipping	dropout=0.1 · weight_decay=1e-4 · layer norm	entropy bonus=0.01 · max_grad_norm=0.5
Train Time	~15 min	~2 hours	~4 hours	~8 hours
Retrain Cadence	Weekly (Sun 22:00 UTC)	Weekly (Sun 22:00 UTC)	Weekly (Sun 22:00 UTC)	Monthly (1st Sunday)
Inference Runtime	CPU · ONNX export	GPU (CUDA) · TorchScript	GPU (CUDA) · TorchScript	CPU · ONNX export
Inference p99	0.3 ms	5.0 ms	8.0 ms	0.5 ms
Explainability	SHAP (TreeExplainer) — per-feature contribution	Attention weights — which bars matter most	Multi-head attention map — long-range dependencies	Action distribution + Q-value — what RL learned
Ensemble Weight	0.35 (highest — primary direction signal)	0.25 (temporal confirmation)	0.25 (context confirmation)	0.15 (management only, not direction)
Strengths	· Tabular feature interactions · Fast inference (µs) · SHAP explainability · Robust to missing features	· Temporal dependencies · Multi-bar momentum patterns · Handles non-stationary sequences · Attention identifies key bars	· Long-range context (120 bars) · Multi-head attention (parallel patterns) · Regime context encoding · Position encoding (time-aware)	· Optimal exit timing · Adaptive position sizing · Learns from delayed rewards · Non-greedy management
Weaknesses	· No temporal awareness · Cannot learn sequences · Tree-based — step functions · Extrapolation poor	· Slower inference (GPU required) · Prone to overfitting on noise · Vanishing gradient on long seq · Less explainable than XGBoost	· Slowest inference (8ms) · Data-hungry (needs 120 bars) · Complex hyperparameter tuning · Position encoding sensitivity	· Sample inefficient (10k episodes) · Reward shaping is difficult · Sim-to-real gap (rew vs live) · Hard to explain decisions
Why In Ensemble?	Best single-model accuracy on tabular features. SHAP provides operator-trustable explanations. Fast inference allows real-time. The "anchor" model.	Captures temporal dynamics that XGBoost cannot (momentum persistence, sequence patterns). Confirms or contradicts XGBoost direction.	Long-range context that LSTM's 60-bar window misses. Multi-head attention captures parallel patterns (e.g., vol regime + trend + session).	Optimal trade management that rule-based systems cannot match. Learns when to hold through noise vs when to trim — adaptive, not fixed rules.

Figure 9.1 — Complete model comparison table with architecture, hyperparameters, strengths, and weaknesses.

Chapter 10 — Integration with TITAN Core

The HAIS integrates with the Strategy Coordinator as a signal source. The AISignal (direction + confidence + RL action + explainability) is emitted on the ZMQ bus. The Strategy Coordinator gates it through the ARDS regime detector before dispatching to the appropriate strategy (trend or mean reversion). The RL management actions are applied to open positions via the Execution Engine. The operator console displays real-time model votes, confidence, and SHAP values.

Signal Flow

- HAIS inference (4 models parallel, ~15ms)
- Ensemble vote (direction + confidence + RL action)
- ZMQ PUB: ai.signal
- Strategy Coordinator
- Regime gate (ARDS)
- If TREND: dispatch to Trend Strategy (Module 5)
- If RANGE: dispatch to Mean Reversion (Module 6)
- If VOLATILE/NEWS: hold, no new entries
- RL action applied to open positions via Execution Engine
- Operator Console (real-time display)

Chapter 11 — Appendix A — Sample AISignal Output

This appendix shows the AISignal for a high-confidence UP signal where all 3 direction models agree and the RL model recommends HOLD (maintain position).

```
{
  "signal_id": "AISIG-2026-06-19-001",
  "timestamp": 1718798400000000000,
  "symbol": "XAUUSD",

  "direction": "UP",
  "confidence": 0.82,
  "probability": { "UP": 0.82, "DOWN": 0.18 },
  "explainability": 0.78,

  "model_votes": {
    "xgboost": { "direction": "UP", "prob": 0.85, "weight": 0.35, "shap_top3": ["ADX",
    "EMA_slope", "Hurst"] },
    "lstm": { "direction": "UP", "prob": 0.78, "weight": 0.25, "attention_peak":
    "bar_-15" },
    "transformer": { "direction": "UP", "prob": 0.80, "weight": 0.25,
    "context_regime": "TREND" },
    "rl": { "action": "HOLD", "q_value": 0.65, "weight": 0.15 }
  },

  "rl_action": "HOLD",
  "rl_q_value": 0.65,

  "top3_features": [
    { "name": "ADX", "shap": +1.85 },
    { "name": "EMA_slope", "shap": +1.20 },
    { "name": "Hurst", "shap": +0.75 }
  ],

  "ensemble_summary": {
    "direction_score": "+0.68 (weighted sum)",
    "agreement": "3/3 unanimous (UP)",
    "confidence_tier": "FULL (conf > 0.65)",
    "position_size_mult": 1.0
  }
}
```

This signal shows a high-confidence UP prediction with unanimous agreement across all 3 direction models. The RL model recommends HOLD (maintain any open position — do not close or trim). The confidence of 0.82 triggers full position sizing (conf > 0.65 threshold). The top-3 SHAP features (ADX, EMA_slope, Hurst) explain 78% of the prediction, providing operator-trustable explainability.